

Francisco Lopes

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EDUCATION

University of Minho

MSc Software Engineering

Braga, Portugal

Sep 2025 – Present

University of Trás-os-Montes and Alto Douro

BSc Software Engineering

Vila Real, Portugal

Sep 2022 – Jun 2025

EXPERIENCE

Software Engineer Intern

Aug 2024 – Sep 2024

Neadvance

Braga, Portugal

- Identified a workflow bottleneck where large Git pushes were difficult to review, and independently designed and built a Git diff visualisation tool from scratch using Python
- Developed a clean interface that parses and displays code changes with clear indentation, highlighting added and removed lines for rapid review
- Delivered a measurable improvement in the team's code review efficiency, with the tool adopted and continued in active use after the internship ended

PROJECTS

Distributed IoT Telemetry Platform | *Java NIO, RxJava, ZeroMQ*

Feb 2026 – Jul 2026

- Designed and implemented concurrent message handling using Java NIO and RxJava reactive streams; ZMQ DEALER/ROUTER for scalable async inter-server communication
- Gained deep practical understanding of distributed coordination, message-passing protocols, and fault-tolerant system design
- MSc Distributed Systems course project — Grade: 18/20

Votium — Blockchain Voting Application | *Solidity, React, JavaScript*

Feb 2025 – Jul 2025

- Architected and deployed smart contracts in Solidity on an Ethereum-compatible (Ganache) network, designed the on-chain data structures for vote storage and verification
- Built the full frontend in React and JavaScript with real-time blockchain event listening, giving me end-to-end ownership of a Web3 product stack
- BSc Final Project — Grade: 17/20

Automated Cloud Deployment & Optimization of AirTrail | *Kubernetes,*

Sep 2025 – Dec 2025

Ansible, Locust

- Automated the full deployment lifecycle of a web application on Google Kubernetes Engine using Ansible, covering cluster provisioning, storage, networking, and health checks
- Load-tested the platform with Locust and used cloud monitoring data to identify and eliminate a scalability bottleneck through autoscaling, cutting the error rate from 21% to 4% and halving latency under load
- MSc Cloud Computing course project — Grade: 17.5/20

TECHNICAL SKILLS

Languages: Python, Java, Bash, JavaScript, Erlang, SQL

Frameworks & Tools: Git, ZeroMQ, RxJava, Java NIO, React, Node.js

Cloud & Infrastructure: Kubernetes (GKE), Docker, Ansible, Google Cloud Monitoring, Locust, PostgreSQL

Domains: Cloud Operations & Monitoring, Distributed Systems, Blockchains, Infrastructure Automation, Developer Tooling

LANGUAGES

Portuguese: Native | **English:** Fluent (written and oral)